

KOHRA ENTERPRISE POLICY INTELLIGENCE

Comprehensive Technical Manual: Prompts, System Instructions & Agent Workflows

1. System & Architecture Overview

The KOHRA Enterprise Policy Intelligence Agent is an enterprise conversational AI platform built to deliver authoritative, zero-hallucination policy intelligence across 30 enterprise repositories (206 policy chunks). The solution enforces post-retrieval Role-Based Access Control (RBAC), automatic temporal supersession resolution, Human-in-the-Loop (HITL) risk gating, multi-format serialization (Prose, JSON, XML, Excel .xlsx, and Email), and real-time millisecond audit logging.

Component	Specification / Tech Stack
System Core	KOHRA Foundations Intelligence Core
Target Architecture	Track 3 Enterprise Conversational AI Agent
LLM Engine	Groq Llama-3.3-70B-Versatile + Local Grounded Synthesizer
Retrieval Engine	BM25 + Semantic Vector Ranking + Reciprocal Rank Fusion (RRF k=60)
Knowledge Base	30 Policy Repositories (9 Official Kohler Docs + 21 Synthetic Governance Repos)
RBAC Governance	6 Roles: EMPLOYEE, MANAGER, HR, FINANCE, LEGAL, ADMIN
Multi-Format Formats	Prose Markdown, JSON, XML, Excel (.xlsx), Handlebars Email

2. Primary LLM System Prompt Specification

The following prompt is injected as the system instruction into the LLM context. It strictly constrains the agent to provided policy chunks, mandates citation extraction, enforces temporal conflict awareness, and requires valid JSON schema output.

```
You are the KOHRA Enterprise Intelligence Agent, an authoritative policy assistant.
Answer the user's question using ONLY the provided CONTEXT DOCUMENTS.

RULES:
1. Ground every statement in the provided documents. Never speculate or hallucinate.
2. If a policy has superseded an older version (e.g. v3.0 replacing v2.0), explicitly state the newer
active terms and note the update.
3. If documents do not contain the answer, say 'I don't have sufficient information about this in the
knowledge base.'
4. Extract precise citations matching the documents you used.
5. You MUST return ONLY a valid JSON object matching this exact schema without markdown wrapping:
{
  "answer": "string (clear, professional markdown text)",
  "confidence": 0.0 to 1.0,
  "grounded": true,
  "citations": [
    {
      "documentTitle": "string",
      "section": "string",
      "version": "string",
      "snippet": "exact brief text snippet from doc",
      "filePath": "relative path to markdown file"
    }
  ]
}
```

```
}
```

3. Context & Prompt Assembly Pipeline

Before sending requests to the LLM, the backend orchestrator formats the retrieved context documents, known policy conflicts/supersessions, conversation history, and user RBAC role into a structured payload:

```
CONTEXT DOCUMENTS:
[DOC 1] Title: "Travel Per Diem Policy" | Section: "Domestic Travel" | Version: "2.1" | Domain: "finance" | File: "policies/finance/travel-per-diem.md"
Daily limit: $75/day for meals and incidentals (breakfast $15, lunch $25, dinner $35)...
---
[DOC 2] Title: "Paid Time Off Policy (Active)" | Section: "Annual Allocation" | Version: "3.0" | Domain: "hr" | File: "policies/hr/pto-policy-v3.md"
Effective January 1, 2026, Kohler enhanced standard annual leave allocation to 20 days PTO...

KNOWN POLICY CONFLICTS / SUPERSESSIONS:
- [SUPERSEDED] Policy "Paid Time Off Policy (Active)" v3.0 supersedes version 2.0 (effective 2026-01-01).

CONVERSATION HISTORY:
User: Hello, I have a travel question.
Assistant: I can assist you with Kohler travel policies.

USER ROLE: EMPLOYEE
USER QUESTION: What's my daily travel per diem for domestic trips?

RESPONSE (JSON only):
```

4. Specialized Agent Workflows & Operations

KOHRA incorporates 11 structured agent workflows designed for end-to-end software development, auditing, and maintenance:

Workflow	Trigger	Operational Scope & Responsibilities
<code>`/orchestrate`</code>	Complex multi-step task	Master agent routing across backend, frontend, security, and design specialists.
<code>`/plan`</code>	Feature planning	4-phase planning methodology: Analysis, Planning (task-slug.md), Solutioning, Implementation.
<code>`/brainstorm`</code>	New feature / vagueness	Socratic Discovery Protocol asking minimum 3 strategic clarification/trade-off questions.
<code>`/create`</code>	Build application	Full-stack creation pipeline coordinating schema, API routes, and Foundations UI components.
<code>`/debug`</code>	Bug or test failure	4-phase systematic debugging: Root cause investigation, traceback verification, evidence fix.
<code>`/test`</code>	Test execution	Executes 8-scenario benchmark suite, unit tests, and Playwright web app tests.
<code>`/deploy`</code>	Production release	5-phase safe deployment: Pre-flight security scan, bundle analysis, zero secrets verification.
<code>`/ui-ux-pro-max`</code>	UI design & review	Enforces Kohler Foundations DS rules, light H1-H6 scales, zero purple ban, accessibility.
<code>`/enhance`</code>	Performance tune	Core Web Vitals profiling, bundle reduction, and React component waterfall elimination.
<code>`/preview`</code>	UI inspection	Generates visual previews and live walkthrough artifacts for UI verification.
<code>`/status`</code>	System check	Audits workspace status, git state, policy chunk counts, and benchmark pass rates.

5. 8 Evaluation Benchmark Scenarios (100% Pass)

The automated verification suite (``npx tsx scripts/test-scenarios.ts``) tests all 8 core enterprise governance scenarios:

#	Scenario	Domain & Role	Target Assertion & Result
1	Single-Domain Grounding	<code>`finance`</code> / EMPLOYEE	Extracts \$75/day per diem; cites Travel Per Diem Policy v2.1. (PASS)
2	Cross-Domain Synthesis	<code>`privacy`</code> + <code>`legal`</code> / LEGAL	Synthesizes DPA encryption + MSA indemnification rules. (PASS)
3	RBAC Containment	<code>`finance`</code> / EMPLOYEE vs FINANCE	EMPLOYEE blocked (3 docs dropped); FINANCE receives full budget schedule. (PASS)
4	Policy Conflict Resolution	<code>`hr`</code> / EMPLOYEE	Flags active v3.0 (20 days) superseding legacy v2.0 (15 days). (PASS)

5	Multi-Format Output	`hr` / ADMIN	Serializes output across Prose, JSON, XML, Excel (.xlsx), and Email. (PASS)
6	Customer Support Terms	`support` / EMPLOYEE	Extracts Kohler India D2C Terms & global warranty terms. (PASS)
7	RBAC Supplier Governance	`legal` / EMPLOYEE vs LEGAL	EMPLOYEE blocked from Supplier Code of Conduct; LEGAL permitted. (PASS)
8	Employee Privacy Notice	`privacy` / EMPLOYEE	Extracts employee data collection rules from official Privacy Notice. (PASS)